Brac University
Department of Computer Science and Engineering

CSE111: Programming Language II Summer 2024 Quiz 2 Marks: 10

Name:	ID:	Section:

1. **Create/Design** a class called **MarksCalculator** such that the results displayed in the output box are generated. [You are not allowed to change the code below]

[10]

Driver Codes/Tester Class:	Outputs:

<pre>public class MarksCalculatorTester {
  public static void main(String[] args) {
    MarksCalculator student1 = new MarksCalculator("Alice");
    System.out.println("Average Marks: " + student1.calculateAverageMark());
    System.out.println("1---------------------------");
    student1.addMarks(85);
    student1.addMarks(92);
    student1.addMarks(78);
    System.out.println(student1.printDetails());
    System.out.println("2---------------------------");
    System.out.println("Average Marks: " + student1.calculateAverageMark());
    System.out.println("3---------------------------");
    student1.updateMarks(99, 1);
    System.out.println(student1.printDetails());
    System.out.println("4---------------------------");
    MarksCalculator student2 = new MarksCalculator("Bob", 80, 90, 88, 94, 99);
    System.out.println("5---------------------------");
    System.out.println(student2.printDetails());
    System.out.println("6---------------------------");
    System.out.println("Average Marks: " + student2.calculateAverageMark());
    MarksCalculator student3 = new MarksCalculator("Sheldon", 79, 85, 98, 90, 100, 96);
    System.out.println(student3.printDetails());
    System.out.println("7---------------------------");
    System.out.println("Average Marks: " + student3.calculateAverageMark());
    student3.addMarks(75);
    student3.updateMarks(88, null);
    student3.updateMarks(55, 4);
    System.out.println(student3.printDetails());
    System.out.println("8---------------------------");
    System.out.println("Average Marks: " + student3.calculateAverageMark());
  }
}</pre>
	<pre>Alice's Details:
	Average Marks: 0.0
	1---------------------------
	Student Name: Alice
	Marks: 85 92 78
	2---------------------------
	Average Marks: 85.0
	3---------------------------
	Student Name: Alice
	Marks: 85 99 78
	4---------------------------
	Bob's Details:
	5---------------------------
	Student Name: Bob
	Marks: 80 90 88 94 99
	6---------------------------
	Average Marks: 90.2
	Sheldon's Details:
	Student Name: Sheldon
	Marks: 79 85 98 90 100 96
	7---------------------------
	Average Marks: 91.33333333333333
	Student Name: Sheldon
	Marks: 79 85 98 90 55 96 75 88
	8---------------------------
	Average Marks: 83.25</pre>

Solution:

```
public class MarksCalculator {
  private String name;
  private int[] marks;
  private int markCount;

  // Constructor
  public MarksCalculator(String name, Integer... info) {
    this.name = name;
    this.marks = new int[100];
    this.markCount = 0;

    if (info.length != 0) {
      for (Integer mark : info) {
        this.marks[markCount] = mark;
        this.markCount++;
      }
    }

    System.out.println(this.name + "'s Details:");
  }

  public void addMarks(int mark) {
    if (markCount < marks.length) {
      this.marks[markCount] = mark;
      this.markCount++;
    } else {
      System.out.println("No more marks can be added.");
    }
  }

  public void updateMarks(int mark, Integer index) {
    if (index == null) {
      this.addMarks(mark);
    } else if (index >= 0 && index < markCount) {
      this.marks[index] = mark;
    } else {
      System.out.println("Invalid index.");
    }
  }

  public double calculateAverageMark() {
    if (this.markCount == 0) {
```

```
      return 0;
    }
    int sum = 0;
    for (int i = 0; i < this.markCount; i++) {
      sum += this.marks[i];
    }
    return (double) sum / this.markCount;
  }

  public String printDetails() {
    String details = "Student Name: " + this.name + "\nMarks: ";
    for (int i = 0; i < this.markCount; i++) {
      if (i > 0) {
        details += " ";
      }
      details += this.marks[i];
    }
    return details;
  }
}
```